

PHYS 150 Notes (Incomplete)

Spring 2026

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Chapter 1

5-12

1.1 Beam Distributions

We have identified a beam matrix

$$\Sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{pmatrix}$$

where $\sigma_{11} = \langle x^2 \rangle$, $\sigma_{12} = \langle xx' \rangle$, $\sigma_{22} = \langle x'^2 \rangle$.

The determinant is defined as

$$\sqrt{\det \Sigma} = \sqrt{\sigma_{11}\sigma_{22} - \sigma_{12}^2}$$

, and is only conserved/constant in linear transport

How do the moments of distribution function evolve with s ? This leads directly to the envelope equation.

Given that $\sigma_x = \sqrt{\langle x^2 \rangle}$, then

$$\begin{aligned} \frac{d\sigma_x}{ds} &= \frac{1}{2\sqrt{\langle x^2 \rangle}} \frac{d}{ds} \langle x^2 \rangle = \frac{1}{2\sigma_x} \frac{d}{ds} \int x^2 F(x, x') dx dx' \\ &= \frac{2}{2\sigma_x} \int xx' F(x, x') dx dx' = \frac{\sigma_{xx'}}{\sigma_x} \end{aligned}$$

The second derivative is

$$\begin{aligned} \frac{d^2\sigma_x}{ds^2} &= \frac{d}{ds} \frac{\sigma_{xx'}}{\sigma_x} = \frac{1}{\sigma_x} \frac{d}{ds} \int xx' F dx dx' - \frac{\sigma_{xx'}}{\sigma_x^2} \frac{d}{ds} \sigma_x \\ &= \frac{1}{\sigma_x} (\sigma_{x'x'} + \langle xx'' \rangle) - \frac{\sigma_{xx'}^2}{\sigma_x^3} \end{aligned}$$

For linear transport, particles satisfy $x'' = -Kx$. So the second derivative now equals

$$\begin{aligned} &= -\frac{K}{\sigma_x} \langle x^2 \rangle + \frac{\sigma_{x'x'}}{\sigma_x} - \frac{\sigma_{xx'}^2}{\sigma_x^3} \\ &= -K\sigma_x + \frac{\sigma_{x'x'}\sigma_x^2 - \sigma_{xx'}^2}{\sigma_x^3} = -K\sigma_x + \frac{\epsilon^2}{\sigma_x^3} \end{aligned}$$

Therefore, the envelope equation is $\sigma_x'' + K\sigma_x = \frac{\epsilon^2}{\sigma_x^3}$, provides an interpretation for ϵ as thermal pressure in beam evolution.

Example 1.1.1 (Drift, $K = 0$)

The envelope equation is

$$\sigma_x'' = \frac{\epsilon^2}{\sigma_x^3}$$

The solution to this equation is

$$\sigma_x = \sigma_{x0} \sqrt{1 + \frac{s^2}{\beta_0^2}}$$

Letting $\sigma_{x0}^2 = \epsilon\beta_0$,

$$\sigma_x = \sigma_{x0} \sqrt{1 + \frac{s^2 \epsilon^2}{\sigma_{x0}^4}}$$

Example 1.1.2 (Constant focusing channel, $K = \text{constant}$, or $K = \kappa_{avg}^2$)

Examples of such are long solenoids or plasma channels. We want to look for solutions of the envelope equation. We can look for equilibrium $\sigma_x'' = 0$, by solving

$$K\sigma_{x,eq} = \frac{\epsilon^2}{\sigma_{x,eq}^3}$$

$$\sigma_{x,eq}^2 = \frac{\epsilon}{\kappa_{avg}}$$

This is a **matching condition**. The envelope of the beam satisfying the matching condition will remain constant.

What happens for small deviations from matching condition? Consider a small deviation from the equilibrium solution, $\sigma_x = \sigma_{x,eq} + \delta\sigma$. Substituting into envelope equation.

$$\sigma_x'' = \sigma_{x,eq}'' + \delta\sigma'' = \delta\sigma''$$

$$\delta\sigma'' + K(\sigma_{x,eq} + \delta\sigma) = \frac{\epsilon^2}{(\sigma_{x,eq} + \delta\sigma)^3}$$

$$\begin{aligned} &= \frac{\epsilon^2}{\sigma_{x,eq}^3 \left(1 + \frac{\delta\sigma}{\sigma_{x,eq}}\right)^3} \\ &\approx \frac{\epsilon^2}{\sigma_{x,eq}^3} \left(1 - 3\frac{\delta\sigma}{\sigma_{x,eq}}\right) \end{aligned}$$

from Taylor expanding the denominator. Since $K\sigma_{x,eq} = \frac{\epsilon^2}{\sigma_{x,eq}^3}$, the equation reduces to

$$\delta\sigma'' + K\delta\sigma + 3\frac{\epsilon^2}{\sigma_{x,eq}^4}\delta\sigma = 0$$

or

$$\delta\sigma'' + 4K\delta\sigma = 0$$

who's solution is a harmonic function with frequency that is twice the frequency of single particles, $\sigma_x = \sqrt{\epsilon\beta}$.

$$\delta\sigma_x = \delta\sigma_{x,0} \cos(2\sqrt{K}s)$$

1.2 Dispersion

So far we considered particle motion at design energy. $p_0 = m_0 c \beta_0 \gamma_0$. In practice, each particle has a different momentum.

$$p = p_0 + \delta p = p_0(1 + \delta)$$

if we define $\delta = \delta p / p_0$ as the relative momentum deviation, and assume $\delta \ll 1$.

Let's see what happens to magnetic rigidity

$$B\rho = \frac{p}{q} = \frac{p_0(1+\delta)}{q} = [B\rho]_0(1+\delta)$$

There are two main effects due to the deviation from reference momentum: bending and focusing.

Example 1.2.1 (Focusing in Quadrupole)

Recall, $f^{-1} = K^2 l_q = \frac{B' l_q}{B\rho}$, so

$$f = \frac{B\rho}{B' l_q} = \frac{[B\rho]_0}{B' l_q} (1+\delta) = f_0(1+\delta)$$

which introduces a **chromatic focusing effect**.

Example 1.2.2 (Bending magnet effect on trajectory)

There is a design trajectory given as $x_d'' = -1/\rho_0$. Weak focusing is given as $x'' + \rho_0^{-2} x = 0$. Recall, $\rho_0 = p_0/(qB_0)$. For particles at difference momenta, $p = p_0(1+\delta)$, $\rho = \rho_0(1+\delta)$. The equation is

$$\begin{aligned} x'' + x_d'' &= -\frac{q}{pv_s} v_s B_0 \\ x'' &= -\frac{q}{p} B_0 \left(1 + \frac{x}{\rho}\right) + \frac{qB_0}{p_0} \\ x'' &= -\frac{q}{p} B_0 \frac{x}{\rho_0}. \end{aligned}$$

Approximating $1/p \approx 1/p_0 - \delta/p_0$,

$$\begin{aligned} x'' &= -\frac{q}{p_0} B_0 \frac{x}{\rho_0} + \delta \frac{qB_0}{p_0} \\ x'' + \frac{1}{\rho_0^2} x &= \frac{\delta}{\rho_0} \end{aligned}$$

which are first order correction for particles not at design energy.

In general, if bending magnet has quadrupole component,

$$x'' + K(s)x = \frac{\delta}{\rho}$$

Chapter 2

5-14

2.1 Dispersion

Recall, dispersion quantifies deviation from reference momentum by $\delta = \Delta p/p \ll 1$. We derived an equation from last lecture for a bending magnet

$$x'' + \frac{1}{\rho^2}x = \frac{\delta}{\rho}$$

The generalization of the bending magnet involves a varying $K(s)$

$$x'' + K(s)x = \frac{\delta}{\rho}.$$

We look for solutions to this equation, $x = x_h + x_p$. In particular, we look for a particular solution proportional to δ since driven term is proportional to δ , $x_p = \delta D$. Plugging in, D must solve

$$D'' + KD = \frac{1}{\rho}.$$

x_p is the contribution to the orbit due to energy deviation. For constant K and ρ , the general solution is

$$D = a \cos \sqrt{K}s + b \sin \sqrt{K}s + \frac{1}{K\rho}.$$

Applying initial conditions, $D(0) = D_0$, $D'(0) = D'_0$, we get

$$D(s) = \frac{1}{K\rho} + \left(D_0 - \frac{1}{K\rho}\right) \cos \sqrt{K}s + \frac{D'_0}{\sqrt{K}} \sin \sqrt{K}s.$$

This solution can be written as a 3x3 matrix.

$$\begin{pmatrix} D(s) \\ D'(s) \\ 1 \end{pmatrix} = \begin{pmatrix} \cos \sqrt{K}s & \frac{1}{\sqrt{K}} \sin \sqrt{K}s & \frac{1 - \cos \sqrt{K}s}{K\rho} \\ \sqrt{K} \sin \sqrt{K}s & \cos \sqrt{K}s & \frac{\sin \sqrt{K}s}{\sqrt{K}\rho} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} D_0 \\ D'_0 \\ 1 \end{pmatrix}$$

Example 2.1.1 (Sector magnet of length l and $\theta \ll 1$)

We know ρ is constant, $K = 1/\rho^2$, and $l = \rho\theta$. The matrix for the sector magnet is

$$\begin{pmatrix} \cos \theta & \rho \sin \theta & \rho(1 - \cos \theta) \\ -\frac{\sin \theta}{\rho} & \cos \theta & \sin \theta \\ 0 & 0 & 1 \end{pmatrix} \approx \begin{pmatrix} 1 & l & \frac{l\theta}{2} \\ 0 & 1 & \theta \\ 0 & 0 & 1 \end{pmatrix}$$

2.1.1 Green's function method for driven linear systems

A second approach to solve the particle offset equation $x'' + K(s) = 1/\rho(s)$, δ offset is normalized into D . General response is obtained by knowing impulse response.

$$F(s) = \int \delta(s - \bar{s}) F(\bar{s}) d\bar{s}$$

Green's function is impulse response, it is solution of

$$G'' + K(s)G = \frac{\delta(s - \bar{s})}{\rho}$$

A particular solution of the original Hill's equation is

$$D_p = \int_0^s d\bar{s} G(s, \bar{s}) \frac{1}{\rho(s)}$$

From the Green's function differential equation, we realize for $s \neq \bar{s}$, the solution is for the homogeneous equation, so the general solution must be a linear combination of solutions of homogeneous equation.

$$G = A(\bar{s})C(s) + B(\bar{s})S(s)$$

Because of continuity, $G(s - \bar{s}) = 0$, and impulse response is $G'(s - \bar{s}) = 1 \Rightarrow A = -S, B = C$, so $G(s, \bar{s}) = S(s)C(\bar{s}) - C(s)S(\bar{s})$. Now that we have G , we can obtain D_p .

$$D_p = \int_0^s d\bar{s} G(s, \bar{s}) \frac{1}{\rho(\bar{s})} = S(s) \int_0^s d\bar{s} \frac{C(\bar{s})}{\rho(\bar{s})} - C(s) \int_0^s d\bar{s} \frac{S(\bar{s})}{\rho(\bar{s})}$$

Example 2.1.2 (Sector magnet)

We know $C(s) = \cos s/\rho$ and $S(s) = \rho \sin \frac{s}{\rho}$, so

$$\begin{aligned} D(s) &= \rho \sin \frac{s}{\rho} \int_0^s d\bar{s} \frac{\cos(\bar{s}/\rho)}{\rho} - \cos \frac{s}{\rho} \int_0^s d\bar{s} \frac{\rho \sin(\bar{s}/\rho)}{\rho} \\ &= \sin \frac{s}{\rho} \rho \sin \frac{\bar{s}}{\rho} \Big|_0^s - \cos \frac{s}{\rho} \rho \cos \frac{\bar{s}}{\rho} \Big|_0^s = \rho (1 - \cos \theta) \end{aligned}$$

Example 2.1.3 (Spectrometer design)

Our design will be a dipole then a drift, we want to maximize dispersion. The transport matrix is

$$\begin{aligned} M &= M_{\text{drift}} M_{\text{bend}} = \begin{pmatrix} 1 & L & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \theta & \rho \sin \theta & \rho (1 - \cos \theta) \\ -\frac{\sin \theta}{\rho} & \cos \theta & \sin \theta \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \cos \theta - \frac{L}{\rho} \sin \theta & \rho \sin \theta + L \cos \theta & \rho (1 - \cos \theta) + L \sin \theta \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix} \end{aligned}$$

Recall, spotsize is related by

$$\sigma_x^2 = M_{11}^2 \sigma_{x_0}^2 - 2M_{11}M_{12} \sigma_{xx'_0} + M_{12}^2 \sigma_{x'_0}^2$$

We simplify to case where at dipole entrance $\sigma_{xx'_0} = 0$ to

$$\sigma_x = \sqrt{M_{11}^2 \sigma_{x_0}^2 + M_{12}^2 \sigma_{x'_0}^2}$$

The dispersion spot size is

$$\sigma_f = D\delta = [\rho (1 - \cos \theta) + L \sin \theta] \delta$$

The resolution of the detector is

$$\delta_{\text{res}} = \frac{\sigma_x}{D} = \frac{\sqrt{\left(\cos \theta - \frac{L}{\rho} \sin \theta\right)^2 \sigma_{x_0}^2 + (\rho \sin \theta + L \sin \theta)^2 \sigma_{x'_0}^2}}{\rho(1 - \cos \theta) + L \sin \theta}$$

In the $L = 0$ case,

$$\delta_{\text{res}} = \frac{\sqrt{\rho^2 \cos^2 \theta \sigma_{x_0}^2 + \rho^2 \sin^2 \theta \sigma_{x'_0}^2}}{\rho(1 - \cos \theta)}$$

And for $\theta = \pi/2$,

$$\delta_{\text{res}} = \sqrt{\sigma_{x'_0}^2} = \sqrt{\frac{\epsilon^2}{\sigma_{x_0}^2}} = \frac{\epsilon}{\sigma_{x_0}}$$

which implies the resolution is limited by emittance

Chapter 3

5-19

3.1 Transverse Dispersion

Let $x = D\delta$ be the displacement with relative momentum δ . We found two methods to solve for D in the last class.

3.2 Achromatic beamlines, $D = 0$

When dispersion is even around midplane S_m , we get $D'(S_m) = 0$. The transport matrix is $M = M_{\text{lens}} M_{\text{drift}} M_{\text{dipole}}$. The condition that $D' = 0$, lets us solve $M_{23} = 0$, which implies $-\frac{\rho}{f}(1 - \cos \theta) + \sin \theta(1 - \frac{L}{f})$, allowing us to solve for $f = L + \rho \tan \frac{\theta}{2}$

3.3 Longitudinal Dispersion

In a longitudinal phase space, the head has higher energy, hits a screen first.

Let τ be the time it takes for particles to travel a beamline of length L . We note there this time depends on the length and the energy of a particle (since higher energy particles tend to travel further distances).

$$\tau = \frac{L}{\beta c}$$

or

$$\tau = \frac{\Delta L}{L} - \frac{\Delta \beta}{\beta}$$

Now the path length is, which depends on offset and angles

$$L = \int dl = \int \sqrt{\left(1 + \frac{x}{\rho}\right)^2 + x^2 + y^2} ds$$

which in the first order, equals

$$\int \sqrt{1 + \frac{2x}{\rho}} dx \int \left(1 + \frac{2x}{\rho}\right) dx$$

Now,

$$\frac{\Delta L}{L_0} = \frac{\int x/\rho dx}{L_0} = \frac{\delta \int D/\rho}{L_0} = \delta \alpha_0$$

where we define α_c is defined as the **momentum compaction factor**

Now, the momentum component of τ . Given that $p = mc\beta\gamma$, and $E^2 = p^2 + m^2$, we get $E\Delta E = p\Delta pc^2$. Then we can obtain that,

$$\frac{\Delta \gamma}{\gamma} = \frac{\Delta E}{E} = \frac{p^2 c^2}{E^2} \frac{\Delta p}{p} = \beta_0^2 \frac{\Delta p}{p}$$

$$\frac{\Delta p}{p} = \frac{\Delta \beta}{\beta} + \frac{\Delta \gamma}{\gamma}$$

$$\frac{\Delta \beta}{\beta} = \frac{\Delta p}{p} - \frac{\Delta \gamma}{\gamma} = \delta - \beta^2 \delta = \frac{1}{\gamma^2} \delta$$

Putting everything together,

$$\frac{\Delta \tau}{\tau} = \left(\alpha_c - \frac{1}{\gamma^2} \right) \delta$$

In storage rings, $\alpha_c - \gamma_+^{-2} = 0$ defines transition energy. Note higher energy particles ($\gamma > \gamma_+$) take longer (have a higher τ). Lower energy particles ($\gamma < \gamma_+$) also take longer. When $\alpha_c - \gamma^{-2} = 0$, it is called an isochronous beamline.

3.4 Slip Factor in Undulators

Essentially, we have an oscillating magnetic field, $B_y = B_0 \sin k_u z$. The trajectory wiggles up and down with a wavelength of $\lambda_u = 2\pi/k_u$. From the lorentz equation,

$$\frac{dp_x}{dz} = -qB_0 \sin k_u z$$

Solving, we get

$$mc\gamma\beta_x = p_x = \frac{qB_0}{k_u} \cos k_u z$$

$$\beta_x = \frac{qB_0}{mck_u\gamma} \cos k_u z = \frac{K}{\gamma} \cos k_u z$$

where $K = \frac{qB_0}{mck_u}$ is the undulator parameter. For a helical undulator, $\beta_y = \frac{K}{\gamma} \sin k_u z$, where

$$\beta_z = \sqrt{1 - \frac{1}{\gamma^2}(\beta_x^2 + \beta_y^2)} = \sqrt{1 - \frac{1 + K^2}{\gamma^2}} \approx 1 - \frac{1 + K^2}{2\gamma^2}$$

Now, we find slip factor. We obtain from letting $z = \beta_z ct$

$$x = \int \beta_x dt = \int \frac{K}{\gamma} \cos(k_u b_z ct) dt$$

$$= \frac{qB_0}{mck_u} \frac{1}{\gamma\beta_z} \frac{1}{ck_u} \sin k_u z = \frac{qB_0^2}{k_u^2} \frac{1}{p} \sin k_u z$$

The disperion is

$$p \frac{\partial x}{\partial p} = D = -\frac{p}{pL} \frac{qB_0}{k_u^2} \sin k_u z = -\frac{1}{p} \frac{qB_0}{k_u^2} \sin k_u z = -\frac{K}{\gamma\beta k_u} \sin k_u z$$

Solving for ρ ,

$$-\frac{1}{\rho} = x'' = -\frac{Kk_u}{\beta\gamma} \sin k_u z$$

The momentum compaction factor is

$$\alpha_c = \frac{1}{L} \int_0^s \frac{D}{\rho} dz = \frac{K^2}{\beta^2 \gamma^2} \langle \sin^2 k_u z \rangle = -\frac{K^2}{2\beta^2 \gamma^2}$$

Now the slip factor is

$$\alpha_c - \frac{1}{\gamma^2} = -\frac{1 + \frac{K^2}{2\beta^2}}{\gamma^2}$$

Chapter 4

5-20

4.1 Motion of Particles off Design Momentum

4.1.1 Longitudinal dispersion

Particles may take different times because they travel different lengths or they have different velocities;

$$\frac{\Delta\tau}{\tau} = \frac{\Delta L}{L} - \frac{\Delta\beta}{\beta} = \left(\alpha_c - \frac{1}{\gamma^2} \right) \delta$$

Recall, we call the **slip factor**, the change in the time of the particle energy as

$$\alpha_c - \gamma^{-2}$$

4.1.2 Magnetic Chicane

Typically, a magnetic chicane is an arrangement of four magnets that create a bump in the design trajectory. The simplest configuration is four different magnets with polarity +, -, -, +, that cause a bump that enter and leave on the same axis. We study this configuration now. Let each of the four magnets be of length L . The design trajectory has length $s = \rho\theta + \rho\theta + \rho\theta + \rho\theta = 4\rho\theta$. Finding momentum compaction factor,

$$\frac{\Delta L}{L} = \frac{\Delta s}{s} = \alpha \frac{\Delta p}{p}$$

since $\rho = p/(qB)$, so ρ is a function of p .

$$\alpha_c = \frac{1}{s} \frac{\partial s}{\partial p} p$$

which we can plug in,

$$\alpha_c = \frac{1}{4\rho\theta} \left(4 \frac{\partial \rho}{\partial p} p \theta + 4\rho \frac{\partial \theta}{\partial p} p \right)$$

Now recall, $\sin \theta = l/\rho = lqB/p$. So $p = lqB/\sin \theta$, or

$$\begin{aligned} \frac{\partial p}{\partial \theta} &= - \frac{lqB \cos \theta}{\sin^2 \theta} \\ &= \frac{1}{4\rho\theta} \left(4 \frac{p}{qB} \theta - 4\rho \frac{p \sin^2 \theta}{lqB \cos \theta} \right) \\ &= \frac{1}{4\rho\theta} (4\rho\theta - 4\rho \tan \theta) = 1 - \frac{\tan \theta}{\theta} \end{aligned}$$

but $\tan \theta > \theta \Rightarrow \alpha_c = 0$. So the slip factor $\alpha_c - \gamma^{-2} < 0$. This means particles of higher energy will take less time to traverse the system. That means if the tail has more energy than the head of the beam, the chicane will compress the beam. This is good for compressing high energy beams.

4.2 Longitudinal Dynamics

Longitudinal dynamics is important when we are accelerating bunches not when we have a continuous beam.

4.2.1 Short Accelerating Gap

Suppose there is an "accelerating" gap where there is an oscillating electric field, $E_z = E(0, z) \sin(\omega t + \phi)$, mimicking a standing wave.

By work-energy theorem, $W = (\gamma - 1)m_e c^2$, the change in kinetic energy is caused by change in kinetic energy

$$\Delta W = \int_{gap} q \vec{E} \cdot d\vec{l} = \int_{-L/2}^{L/2} E_z(0, z) \sin(\omega t(z) + \phi) dz$$

where

$$t(z) = \int \frac{dz}{v_z(z)}$$

We can write ΔW as an alternative definition to become the Ponofsky equation

$$\Delta W \equiv_{\text{Ponofsky equation}} q V_0 T \sin \phi$$

where we define

$$V_0 \equiv_{\text{cavity voltage}} \int_{-L/2}^{L/2} E_z(0, z) dz$$

$$T \equiv \frac{1}{V_0} \left[\int_{-L/2}^{L/2} E_z(0, z) \cos(\omega t) dz + \frac{\cos \phi}{\sin \phi} \int_{-L/2}^{L/2} E_z(0, z) \sin(\omega t) dz \right]$$

where T is the **transit time factor**. All the physics is hidden in T . Usually $E_z(0, z)$ is an even function, symmetric about $z = 0$. For a very short cavity, v is constant, so $t(z) = z/v$. For a RF beam in this short cavity approximation

$$\omega t = \frac{2\pi}{T_{RF}} \frac{z}{\beta c} = \frac{2\pi}{\lambda_{RF}} \frac{z}{\beta}$$

Substituting in expression for T ,

$$T = \frac{1}{V_0} \left[\int E_z(0, z) \cos \frac{2\pi z}{\lambda_{RF} \beta} + \cot \phi \int_{-L/2}^{L/2} E_z(0, z) \sin \frac{2\pi z}{\lambda_{RF} \beta} \right]$$

by parity rules, the second term is 0, so T becomes

$$T = \frac{\int E_z(0, z) \cos \frac{2\pi z}{\lambda_{RF} \beta} dz}{\int E_z(0, z) dz} \leq 1$$

Example 4.2.1 (Pillbox Cavity with Constant Field across gap g)

We first compute T

$$T = \frac{\int_{-g/2}^{g/2} E_0 \cos \frac{2\pi z}{\lambda_{RF} \beta} dz}{E_0 g} = \frac{\sin \frac{\pi g}{\beta \lambda_{RF}}}{\frac{\pi g}{\lambda_{RF} \beta}}$$

Use of short RF gaps in combination with longitudinal dispersion as temporal lenses for bunched beams, which can eventually be used for compression

$$\Delta \gamma = \frac{q V_0 T \sin \phi}{m c^2}$$

We approximate the phase to be approximately 1; $\phi \ll 1$. After the gap,

$$\gamma_f = \gamma_0 + \frac{q V_0 T \phi}{m c^2}$$

$$\delta = \frac{\Delta p}{p} = \frac{1}{\beta^2} \frac{\Delta \gamma}{\gamma} = \frac{qV_0 T \phi}{mc^2 \beta^2 \gamma}$$

where $\phi = \omega \tau_0$, where τ_0 is the arrival time of particles at the center of the gap with respect to reference particle. For the momentum

$$\delta_f = \delta_0 + 2\omega \tau_0$$

which is a linear approximation

$$\begin{aligned} \tau_f &= \tau_0 \\ \begin{pmatrix} \tau \\ \delta \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ 2\omega & 1 \end{pmatrix} \begin{pmatrix} \tau_0 \\ \delta_0 \end{pmatrix} \end{aligned} \quad (4.1)$$

where the 2x2 matrix is the effect of the short accelerating gap, resembling a transport matrix of lens.

In the absence of dispersion, with just a drift,

$$\frac{\Delta \tau}{\tau} = \frac{\Delta L}{L} - \frac{\Delta \beta}{\beta} = \left(\alpha_c - \frac{1}{\delta^2} \right) \delta$$

we know $\Delta L/L = 0$ and $\alpha_c = 0$, so

$$\Delta \tau = -\frac{1}{\gamma'^2} \tau \delta$$

As $\Delta \delta = 0 \Rightarrow \delta_f = \delta_0$ The linear transport in longitudinal space for a drift is therefore

$$\begin{pmatrix} \tau_f \\ \delta_f \end{pmatrix} = \begin{pmatrix} 1 & -\frac{L}{\beta c \gamma'^2} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \tau_0 \\ \delta \end{pmatrix} \quad (4.2)$$

The combined action will be

$$M = \begin{pmatrix} 1 & -\frac{L}{\beta c \gamma'^2} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \alpha \omega & 1 \end{pmatrix} = \begin{pmatrix} 1 - \frac{L \alpha \omega}{\beta c \gamma'^2} & -\frac{L}{\beta c \gamma'^2} \\ \alpha \omega & 1 \end{pmatrix}$$

And so longitudinal focus occurs

$$L = f = \frac{\beta c \gamma'^2}{\alpha \omega} = \frac{\beta^3 \gamma^3 m c^3}{q V_0 T \omega}$$

which works well for an RF cavity or when there is a lot of voltage in the cavity. A cavity is useful for beam compression.

Chapter 5

5-21

5.1 Hamiltonian Mechanics

We define a function in terms of a coordinate and momenta, $H(q, p)$. The equations of motion are

$$\dot{q} = \frac{\partial H}{\partial p}$$
$$\dot{p} = -\frac{\partial H}{\partial q}$$

In simple systems,

$$H = \frac{p^2}{2m} + U(q)$$

in which the equation of motion is what we know already

$$\dot{q} = p/m$$
$$\dot{p} = -\frac{\partial U}{\partial q}$$

The advantages of Hamiltonian formulation:

- If H is time independent, H is conserved quantity of motion
- Symmetries and conserved quantities are more evident
- Phase space picture
- Liouville's Theorem
- Symplectic structure for Hamiltonian flow ($\det M = 1$)
- Well-defined procedure to change variables, canonical transformation

For a non relativistic free particle, $H = p^2/2m$ and the equations of motion are

$$\dot{p} = 0$$
$$\dot{q} = p/m$$

For a relativistic free particle, $H = \sqrt{p^2 c^2 + m^2 c^4}$. For a relativistic charged particle interacting with an electric field

$$H = \sqrt{(\vec{p}_c - q\vec{A})^2 c^2 + m^2 c^4} + q\phi$$

where

$$\vec{B} = \nabla \times \vec{A}$$

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and

$$\vec{E} = -\nabla\phi - \frac{\partial \vec{A}}{\partial t}$$

Note: $\vec{p}_c$ in Hamiltonian might not be the mechanical momentum, $\vec{p} = mc\gamma\vec{\beta}$. In the above Hamiltonian

$$\vec{p}_c = \vec{p} + q\vec{A}$$

H is usually the energy of particles, not always.

We now use this formalism to describe acceleration of particles by an EM wave. First, vector potential can be described as one component in a 1D electromagnetic wave

$$A_z = -\frac{E_0}{k_z v_\phi} \cos(k_z(z - v_\phi t))$$

so

$$E_z = -\frac{\partial A_z}{\partial t} = E_0 \sin k_z(z - v_\phi t)$$

In a vacuum, phase velocity $v_\phi = c$, can be different in a waveguide or structure The 1D Hamiltonian becomes

$$H(z, p_{z,c}) = \sqrt{(p_{z,c} - qA_z)^2 c^2 + m^2 c^4}$$

The equations of motion

$$\begin{aligned} \dot{p} &= -\frac{\partial H}{\partial z} \\ \dot{z} &= \frac{\partial H}{\partial p_{z,c}} = \frac{2c^2(p_{z,c} - qA_z)}{2\sqrt{(p_{z,c} - qA_z)^2 c^2 + m^2 c^4}} = \frac{m\gamma v_z c^2}{mc^2\gamma} = v_z \end{aligned}$$

Changing variables, $\xi = z - v_\phi t$ that is comoving with wave,

$$\begin{aligned} \bar{H}(\xi, p_{\xi,c}) &= H - v_\phi p_{z,c} \\ &= \sqrt{(p_{\xi,c} - qA_z)^2 c^2 + m^2 c^4} - v_\phi p_{\xi,c} \\ &= \sqrt{\left(p_{\xi,c} + q\frac{E_0}{k_z v_\phi} \cos k_z \xi\right)^2 + m^2 c^4} - v_\phi p_{\xi,c} \\ \dot{\xi} &= \frac{\partial \bar{H}}{\partial p_{\xi,c}} = v_z - v_\phi \\ \dot{p}_{\xi,c} &= -\frac{\partial \bar{H}}{\partial \xi} = \frac{v_z}{v_\phi} q E_0 \sin k_z \xi \end{aligned}$$

The derivative of canonical momentum

$$\frac{d(p_\xi = p_{\xi,c} - qA_z)}{dt} = \dot{p}_{\xi,c} - q\frac{d}{dt}A_z = \dot{p}_{\xi,c} - q\frac{\partial A_z}{\partial t} - qv_z\frac{\partial A_z}{\partial z} = -q\frac{\partial A_z}{\partial t} = qE_z$$

Let us rewrite $\bar{H}$ in terms of mechanical quantities (p_ξ, β, γ)

$$\bar{H} = \sqrt{p_\xi^2 c^2 + m^2 c^4} - v_\phi p_\xi + qv_\phi \frac{E_0}{k_z v_\phi} \cos k_z \xi$$

where the transformation is

$$p_{\xi,c} = p_\xi + qA_z$$

and $p_\xi = mc\beta\gamma$. So the Hamiltonian becomes

$$\bar{H} = mc^2 \sqrt{\beta^2 \gamma^2 + 1} - mc^2 \frac{v_\phi}{c} \beta \gamma + mc^2 \alpha \cos k\xi$$

defining $\alpha = qE_z/(mc^2 k_z)$ and $\beta_\phi = v_\phi/c$, so

$$\bar{H} = \sqrt{\beta^2 \gamma^2 + 1} - \beta_\phi \beta \gamma + \alpha \cos k\xi$$

The interaction depends on phase velocity (β_ϕ) and normalized amplitude (α), which are unitless parameters. There are two cases to look at. For $\alpha > 1$, we have violent acceleration. For $\alpha < 1$, gentle acceleration. α describes the amount of energy acquired in one period. In violent acceleration, particles become relativistic in less than a period.

5.1.1 Violent Acceleration

In relativistic waves $v_\phi = c$ to avoid dephasing

$$\begin{aligned}\frac{H}{mc^2} &= \sqrt{\beta^2\gamma^2 + 1} - \beta\gamma + \alpha \cos k\xi = \gamma - \beta\gamma + \alpha \cos k\xi \\ &= \frac{\gamma(1 - \beta)}{\sqrt{1 - \beta^2}} + \alpha \cos k\xi \\ &= \sqrt{\frac{1 - \beta}{1 + \beta}} + \alpha \cos k\xi\end{aligned}$$

Define the **rapidity** χ , to be

$$\chi = \sqrt{\frac{1 + \beta}{1 - \beta}}$$

For $\beta = 0$, $\chi = 1$. For $\beta \rightarrow 1$, $\chi \rightarrow \infty$.

$$\frac{H}{mc^2} = \frac{1}{\chi} + \alpha \cos k\xi$$

The particle moves along constant Hamiltonian contours. For $\chi \rightarrow \infty$, phase becomes stationary, no more phase slippage between particle and waves

$$H = \alpha \cos k\xi = \alpha \cos \phi_f$$

We also obtain a **trapping threshold**. There are some particles that pick up some energy but get dephased so they lose that energy again.

Chapter 6

5-26

We now study the interaction of charged particles with a wave of amplitude E_0 and wavenumber k in the wave-moving reference frame, $\zeta = z - v_\phi t$ where v_ϕ is phase velocity. Recall, we used Hamiltonian formalism with Hamiltonian

$$H = \sqrt{p_\zeta^2 c^2 + m^2 c^4} - v_\phi p_\zeta + \frac{qE_0}{x} \cos k\zeta$$

In the case of violent acceleration, in speed of light waves, $v_\phi = c$.

$$\frac{H}{mc^2} = \chi^{-1} + \alpha \cos k\zeta$$

where

$$\alpha = \frac{qE_0}{mck}$$

is the **normalized wave amplitude** and

$$\chi = \sqrt{\frac{1+\beta}{1-\beta}}$$

is the **rapidity**. Note $\chi = 1$ for $\beta = 0$ and approaches ∞ when $\beta \rightarrow 1$. As particles become highly relativistic, $\chi^{-1} \rightarrow 0$ and $H \approx \alpha \cos \phi_f = \text{constant} \Rightarrow$ particles stop slipping in the wave

When particles interact at rest $\chi = 1$, then

$$H = 1 + \alpha \cos \phi_0$$

For trapping, we want to know if these particles can be accelerated to very high energy. A **barely trapped particle** has $\phi_f = 0$.

$$\frac{H_f}{mc^2} = \chi^{-1} + \alpha \cos \phi_f = \alpha$$

In the worst case scenario $\phi_0 = -\pi$

$$\frac{H_i}{mc^2} = 1 + \alpha \cos \phi_0 = 1 - \alpha$$

so $H_i = H_f$ implies that

$$\alpha = \frac{1}{2}$$

is the **trapping threshold** for a stationary particle

For a good quality beam, we want the final phase to be $\phi_f = \pi/2$. or $H_f/(mc^2) = 0$ which implies that $\alpha = 1$. Recall, for radio frequency (S-band) guns, $f = 3\text{GHz}$, $\lambda = 10\text{cm}$, $E_{tt} = 18\text{MV/m}$. To design a THz gun, $f = 0.3\text{THz}$, $\lambda = 1\text{mm}$, $E_{tt} = 1.8\text{GV/m}$.

In the **Gentle acceleration** case, $\alpha < 1$. We start with constant velocity approximation, equal to phase velocity. We call the synchronous particle to be the particle traveling with the wave $\beta = v_\phi$. Expand H around resonant particle that has momentum $p_r = mc\beta_\phi\gamma_\phi = mv_\phi\gamma_\phi$ to follow longitudinal dynamics. In general, we write

$$p = p_r + \delta p$$

$$\frac{H}{mc^2} = \sqrt{1 + \frac{(p_r + \delta p)^2}{(mc)^2}} - \frac{v_\phi(p_r + \delta p)}{mc^2} + \alpha \cos k\zeta$$

We linearize and look at particles close to resonant momentum $\delta p \ll p_r$. The Hamiltonian is

$$\begin{aligned} \frac{H}{mc^2} &= \sqrt{\left(1 + \frac{p_r^2}{mc^2}\right) \left(1 + \frac{2\delta p}{p_r} + \frac{(\delta p)^2}{p_r^2}\right)} - \frac{v_\phi(p_r + \delta p)}{mc^2} + \alpha \cos k\zeta \\ &= \frac{(\delta p)^2}{2(mc)^2\gamma^3} + \alpha(\cos k\zeta + 1) \end{aligned}$$

From this new Hamiltonian $H(\zeta, \delta p)$, the equation of motion are

$$\dot{\zeta} = \frac{\partial H}{\partial \delta p} = \frac{\delta p}{m\gamma^3}$$

$$\delta \dot{p} = -\frac{\partial H}{\partial \zeta} = mc^2 \alpha k \sin k\zeta = qE_0 \sin k\zeta$$

This gives rise to "pendulum-like dynamics". $m\gamma^3$ represents a longitudinal relativistic mass increase.

Phase space is clearly separated in stable and unstable regions by separatrix. There is an equilibrium point at $\delta p = 0$ and $k\zeta = \pi$. There is also small amplitude oscillation around equilibrium (linearized potential).

$$H_{\text{small}} = mc^2 \left[\frac{(\delta p)^2}{2(mc)^2\gamma^3} + \alpha \frac{(k\zeta)^2}{2} \right]$$

Now the equations of motion for small amplitude become

$$\delta \dot{p} = -\frac{\partial H}{\partial \zeta} = -\alpha k^2 mc^2 \zeta$$

$$\dot{\zeta} = \frac{\partial H}{\partial p} = \frac{\delta p}{m\gamma^3}$$

$$\ddot{\zeta} = -\frac{\alpha k^2}{\gamma^3} \zeta$$

which is a harmonic oscillator with synchrotron frequency

$$\omega_s = \sqrt{\frac{\alpha}{\gamma^3}} kc \ll \omega$$

Typically, the synchrotron frequency is much smaller than driving wave frequency. We now find the separatrix. At $k\zeta = 0, \pi$ and $\delta p = 0$, then at the separatrix, the Hamiltonian has the constant value,

$$\frac{H}{mc^2} = 2\alpha$$

then

$$\frac{\delta p}{p} = \pm \sqrt{\frac{2\alpha\gamma_r}{\beta_r^2} (1 - \cos k\zeta)} = \pm \sqrt{\frac{4\alpha\gamma_r}{\beta_r^2}} \sin \frac{k\zeta}{2}$$

In order to accelerate in stable manner with $\alpha < 1$, need to change v_ϕ along acceleration to taper, $v_\phi = \omega/k$. By changing ω , change spatial periodicity.

Chapter 7

5-28

There exist special, resonant energy or momentum for which $\beta_r = \beta_\phi = v_\phi/c$ is synchronous with the wave. There is pendulum-like motion

Recall last chapter, this type of acceleration has limited energy gain (especially for low α). The solution is to change v_ϕ or γ_r to sustain large particle acceleration. Typically in linac, we typically change spatial periodicity k , but in synchrotron, we change ω for $v_\phi = \omega/k$.

$$\frac{d\gamma_r}{dz} = \frac{qE_0 \sin \phi_r}{m_0 c^2}$$

where we define ϕ_r as a **resonant phase**. We can't change ϕ_r too fast or particles can't keep up. We have to modulate it slowly, so we parameterize tapering. For $\phi_r = \pi/2$, we have maximal gradient. For $\phi_r = 0$, no acceleration (stationary bucket). With these definitions, we can now define a **resonant particle** with resonant phase and resonant energy. Note, we have to build the structure that allows this "resonant" acceleration profile.

We now ask what happens to particles with coordinates close to resonant particles.

$$\begin{aligned}\delta p &= p - p_r \\ \dot{p}_r &= mc^2 k \alpha \sin \phi_r \\ \delta \dot{p} &= mc^2 k \alpha (\sin \phi - \sin \phi_r)\end{aligned}$$

which is pendulum like motion. Recall in Hamiltonian,

$$\frac{V(\phi)}{mc^2} = \alpha(\cos \phi + 1) + \alpha \phi \sin \phi_r$$

where we add the second term to ensure the correct equation of motion in the equation before,

$$-\frac{\partial V}{\partial \phi} = \alpha \sin \phi - \alpha \sin \phi_r$$

We can calculate equilibrium points,

$$\frac{\partial V}{\partial \phi} = 0 \Rightarrow \sin \phi = \sin \phi_r$$

$$\phi = \phi_r$$

is a minimum

$$\phi = \pi - \phi_r$$

is a maximum. We study motion around equilibrium $\delta\phi = \phi - \phi_r$.

$$\cos \phi + \phi \sin \phi_r \approx \cos(\phi_r + \delta\phi) + \phi_r \sin \phi_r + \delta\phi \sin \phi_r$$

$$\approx \cos \phi_r + \frac{\delta\phi^2}{2} \cos \phi_r + \phi_r \sin \phi_r$$

which is parabolic around equilibrium point. So the Hamiltonian around equilibrium point is,

$$\frac{H}{mc^2} = \frac{\beta_r^2}{2\gamma_r} \left(\frac{\delta p}{p_r} \right)^2 + \alpha \frac{\delta \phi^2}{2} \cos \phi_r$$

Stationary states are,

$$k_s = \sqrt{\frac{\alpha \gamma_r}{\beta_r^2}} \cos \phi_r$$

Moving bucket in phase space will become smaller (smaller region of stability). We now compute the separatrix to show this

$$\begin{aligned} \frac{H_{sep}}{mc^2} &= V(\pi - \phi_r) = \alpha [\cos(\pi - \phi_r) + (\pi - \phi_r) \sin(\pi - \phi_r) + 1] \\ &= \alpha [-\cos \phi_r + (\pi - \phi_r) \sin \phi_r + 1] \end{aligned}$$

Recall,

$$\frac{H}{mc^2} = \frac{\beta_r^2}{2\gamma_r} \left(\frac{\delta p}{p_r} \right)^2 + \alpha V(\phi) = \frac{H_{sep}}{mc^2}$$

To solve for $\delta p/p$ and find separatrix equation for moving bucket

$$\frac{\delta p}{p} = \sqrt{\frac{2\gamma_r \alpha}{\beta_r^2}} [-\cos \phi - \phi \sin \phi_r - 1 + (1 - \cos \phi_r) + (\pi - \phi_r) \sin \phi_r]$$

The max of $\delta p/p$ is at ϕ_r , so we substitute $\phi = \phi_r$

$$\frac{\delta p}{p}(\phi_r) = \sqrt{\frac{2\gamma_r \alpha}{\beta_r^2}} [2 \cos \phi_r + (\pi - 2\phi_r) \sin \phi_r]$$

for $\phi_r = \pi/2$, $\delta p/p|_{max} \rightarrow 0$, no more phase stability. In design of an acceleration, there is a compromise between gradient and extent of stable region. We move the buckets by a constant resonant phase that can be described by moving bucket Hamiltonian. We certainly can design accelerator for example with a resonant phase $\phi_r(z)$, a non constant phase, so now we have a time-dependent $H(z)$. We can study these with **adiabatic invariant** which is the area of moving bucket.

7.1 Effect of Acceleration on Transverse Dynamics

Recall, we can describe transverse dynamics with Hill's Equation

$$x'' + Kx = 0$$

K for a quadrupole $= \frac{B'}{B\rho}$. Recall we derived with Lorentz force

$$\frac{dp_x}{dt} = -qv_z B_y = -qv_z B' x$$

$$(\beta \gamma x)' + \frac{qB}{mc} x = 0$$

If $\beta \gamma$ is constant, we get Hill's equation. If there is acceleration, $\beta \gamma$ is not constant

$$(\beta \gamma)' x' + \beta \gamma x'' + \frac{qB}{mc} x = 0$$

$$x'' + \frac{(\beta \gamma)'}{\beta \gamma} x' + \frac{qB'}{mc \beta \gamma} x = 0$$

The effect of acceleration looks like a damping term. We can generalize Courant-Snyder invariant to account damping by multiplying by $\beta \gamma$.

$$\epsilon \rightarrow \beta \gamma \epsilon = \epsilon_N \equiv \text{normalized emittance}$$

Let's see how the envelope equation changes

$$\sigma_x = \sqrt{\langle x^2 \rangle}$$

Recall,

$$\sigma_x'' = \frac{\langle x x'' \rangle}{\sigma_x} + \frac{\epsilon^2}{\sigma_x^3}$$

$$x'' = -\frac{(\beta\gamma)'}{\beta\gamma} - kx$$

$$\sigma_x'' = -\frac{(\beta\gamma)'}{\beta\gamma} \frac{\langle x x' \rangle}{\sigma_x} - K \frac{\langle x^2 \rangle}{\sigma_x} + \frac{\epsilon^2}{\sigma_x^3}$$

We have the relations

$$\langle x x' \rangle = \sigma_{x'}$$

$$(\beta\gamma)' = \frac{\gamma'}{\beta}$$

Then

$$-\frac{\gamma'}{\beta^2\gamma} \frac{\sigma_{x'}}{\sigma_x} - K\sigma_x + \frac{\epsilon^2}{\sigma_x^3}$$

which goes back to non acceleration case for $\gamma' = 0$.

Chapter 8

6-2

8.1 Smooth space charge

For an infinitely long cylindrical beam of radius r with current at speed v

$$E_r = \frac{Ir}{2\pi\epsilon_0 a^2 v}$$

The magnetic field is given by Ampere's Law

$$B_\phi = \frac{\mu_0 Ir}{2\pi a^2}$$

The force is

$$\begin{aligned} F_r &= q(E_r - vB_\phi) = q \frac{I}{2\pi a^2 r} r \left(\frac{1}{\epsilon_0 v} - v\mu_0 \right) \\ &= q \frac{Ir}{2\pi a^2 \epsilon_0 v} (1 - v^2 \epsilon_0 \mu_0) = q \frac{Ir}{2\pi a^2 \epsilon_0 v} \frac{1}{\gamma^2} \end{aligned}$$

Space charge forces are neglected at high energy $\gamma \gg 1$. Only important at the beginning of acceleration. For these infinitely long, uniform beams

Since F_r is linear in r , then

$$r'' = \frac{K}{a^2} r$$

where K is the **beam perveance**,

$$\begin{aligned} K &\equiv \frac{qI}{2\pi\epsilon_0 \beta c \gamma^2 \beta^2 \gamma m c} \\ &= \frac{qI}{2\pi\epsilon_0 m c^2} \frac{1}{\gamma^2 \beta^2} = \frac{2I}{I_A} \frac{1}{\beta^3 \gamma^3} \end{aligned}$$

where I_A is the **Alfven current**.

$$I_A = \frac{4\pi\epsilon_0 m c^2}{q}$$

Everything above was done for a single particle, we now move to envelope equation

8.2 Radial Envelope Equation

There is no acceleration. Recall,

$$\sigma'' + K_{\text{foc}} \sigma = \frac{\epsilon^2}{\sigma_3} = \frac{\epsilon_n^2}{(\beta\gamma)^2 \sigma_3}$$

Adding space charge

$$\sigma'' + K_{\text{foc}} \sigma = \frac{\epsilon_n^2}{(\beta\gamma)^2 \sigma_3} + \frac{2I}{I_A \sigma (\beta\gamma)^3}$$

Where the first term on LHS is emittance and the second term is the space charge term. Comparing the two terms allows us to determine if dynamics is space charge dominated or not.

$$\rho = \frac{\text{space charge term}}{\text{emittance term}} = \frac{\frac{K}{\sigma}}{\frac{\epsilon_n^2}{(\beta\gamma)^2} \frac{1}{\sigma^3}} = \frac{2I}{I_A} \frac{\sigma^2}{\beta\gamma\epsilon_n^2}$$

If $\rho \gg 1$, then space charge dominates. If $\rho \ll 1$, then emittance dominates. Note σ^2 dependence, for tightly focused beam (small σ), the angle is so tight that space charge has no time to affect envelope.

8.2.1 Space charge dominated beam transport

$$\sigma'' + K\sigma = \frac{2I}{I_A(\beta\gamma)^3} \frac{1}{\sigma}$$

Let's look for equilibrium solution, $\sigma'' = 0$.

$$\sigma_{eq} = \sqrt{\frac{2I}{I_A(\beta\gamma)^3} \frac{1}{K}}$$

called **Brillouin flow**. What happens for small deviations from equilibrium?

$$\sigma = \sigma_{eq} + \delta\sigma$$

We get

$$\delta\sigma'' + K(\sigma_{eq} + \delta\sigma) = \frac{2I}{I_A(\beta\gamma)^3} \frac{1}{\sigma_{eq} + \delta\sigma} = \frac{2I}{I_A(\beta\gamma)^3} \frac{1}{\sigma_{eq}} (1 - \delta\sigma/\sigma_{eq})$$

which simplifies to

$$\delta\sigma'' + 2K\delta\sigma = 0$$

$\Rightarrow$ Brillouin flow is stable equilibrium where oscillations around equilibrium has frequency

$$\omega = \sqrt{2K} = \sqrt{\frac{4I}{I_A(\beta\gamma)^3 \sigma_{eq}^2}} = \sqrt{\frac{e^2 n}{m\epsilon_0(\beta\gamma)^3}} = \omega_p$$

where ω_p is the **plasma frequency**.

Going beyond infinitely long beam approximation (real beam). Mismatch slice oscillation causes emittance growth

Emittance growth due to non linear space charge forces.

If ρ is uniform, and beams are infinitely long, then fields are linear. If we calculate E field for non uniform distribution, there is some non-linearity. RMS-emittance will grow due to non nonlinearities, but leads to irreversible phase space degradation, wave breaking